Gradient Descent

The optimizer updates weights with the learning rate on each training step, and the loss shrinks until convergence appears. A batch of samples drives each weights update during an epoch so noisy gradients still move toward a useful minimum of loss. **Momentum** smooths each step, while weights decay limits overfitting. See x=1 and 2024 for the version we use in class.

Backpropagation

Neural networks compute gradients through the chain rule, and each layer passes activations forward before errors flow back. Weights change after gradients reach the input layer of a model and the loss guides how far each update should move weights. **Dropout** randomly silences neurons, which improves generalization.

Tie Breaking

alpha bravo charlie delta echo foxtrot golf hotel india juliet kilo lima mike november oscar papa quebec romeo sierra tango uniform victor whiskey xray yankee zulu amber coral ivory jade

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역전파를 이해하면 신경망이 보인다